

Tool Hub

Hybrid Architecture — FastAPI + NuxtUI +
LangChain + Ollama

Tài liệu kỹ thuật dự án

Phiên bản 1.0

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Công nghệ: FastAPI, NuxtUI/Vue 3, LangChain, Ollama/Qwen3
Triển khai: Docker Compose (local), Vercel + Render (cloud)

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Tong quan du an

Mo ta

Tool Hub la mot he thong web tích hop nhieu cong cu phan tich du lieu (tools) duoi dang plugin, duoc xay dung theo kien truc hybrid. Tool dau tien trong he thong la Data Analytics Agent — mot agent thong minh su dung mo hinh ngon ngu lon (LLM) de phan tich du lieu tu file CSV, dua ra thong ke, bieu do, va bao cao tu nhien.

Muc tieu

- Xay dung mot Tool Hub co kha nang mo rong bang co che plugin (decorator).
- Tích hợp agent phân tích dữ liệu với 6 công cụ chuyên sâu.
- Hỗ trợ đa bảng (multi-table) — agent có thể làm việc với nhiều file CSV trong cùng một dự án.
- Trả lời bằng cả tiếng Việt và tiếng Anh.
- Triển khai bằng Docker Compose (local) hoặc Vercel + Render (cloud).

Cong nghe chinh

- **FastAPI** — Backend REST API (Python 3.12).
- **NuxtUI / Vue 3** — Frontend SPA (Node 20).
- **LangChain 0.3** — Agent orchestration, ReAct loop.
- **Ollama + Qwen3:1.7b** — LLM inference (CPU).
- **Pandas + Plotly** — Xử lý dữ liệu và vẽ biểu đồ.
- **ECharts (Vue-ECharts)** — Hiển thị biểu đồ ở frontend.
- **Docker Compose** — Đóng gói 3 dịch vụ.

Cau truc thu muc

So do cay

Duoi day la cau truc day du cua du an (chi bao gom file nguon, loai bo node_modules, .nuft, .output, va __pycache__):

```
1 tool-hub/|—
2 .gitignore|—
3 README.md|—
4 docker-compose.yml|—
```

```

5 render.yaml|—
6 vercel.json|—
7
8 ollama/|—
9   Dockerfile|—
10  entrypoint.sh|—
11
12 backend/|—
13   .dockerignore|—
14   .env|—
15   Dockerfile|—
16   requirements.txt|—
17   sample_data/|—
18     sales_sample.csv|—
19   app/|—
20     __init__.py|—
21     config.py|—
22     main.py|—
23     core/|—
24       __init__.py|—
25       registry.py|—
26     tools/|—
27       __init__.py|—
28       tool_template/|—
29         __init__.py|—
30         router.py|—
31       data_analytics/|—
32         __init__.py|—
33         router.py|—
34         schemas.py|—
35         project_manager.py|—
36         agent/|—
37           __init__.py|—
38           analyst.py|—
39           prompts.py|—
40         analyst_tools/|—
41           __init__.py|—
42           chart_generator.py|—
43           csv_reader.py|—
44           query_data.py|—
45           statistics.py|—
46     uploads/|—
47       .gitkeep|—
48
49 frontend/|—
50   .dockerignore|—
51   Dockerfile|—
52   nuxt.config.ts|—
53   package.json|—
54   tsconfig.json|—

```

```
55  app.vue|—
56  layouts/|└─
57    default.vue|—
58  pages/||—
59    index.vue|└─
60    tools/|└─
61      data-analytics.vue|—
62  composables/|└─
63    useApi.ts|—
64  components/||—
65    hub/|||—
66      Sidebar.vue||└─
67      ToolCard.vue|└─
68    data-analytics/||—
69      ChartView.vue|└─
70      FileUpload.vue|└─
71      LLMReport.vue|└─
72      MultiFileUpload.vue|└─
73      ProjectPanel.vue|└─
74      StatsTable.vue|└─
75      TableList.vue└─
76  public/
```

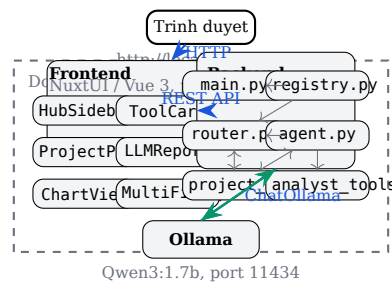
Giai thich tung module

Module	Chuc nang
ollama/	Custom Docker im- age tu dong pull model Qwen3 khi khoi dong
backend/app/config.py	Cau hinh Pydantic (model, up- load, CORS)
backend/app/main.py	FastAPI app, CORS middle- ware, lifes- pan auto- discovery
backend/app/core/registry.py	Plugin reg- istry — @register_tool() decorator
backend/app/tools/	Thu muc chua cac tool plugin
backend/app/tools/tool_template/	Mau de tao tool moi
backend/app/tools/data_analytics/	Tool Data Analytics (chinh)
backend/app/tools/data_analytics/router.py	9 endpoint REST API
backend/app/tools/data_analytics/schemas.py	Pydantic models (Re- quest/Re- sponse)
backend/app/tools/data_analytics/project_manager.py	Quan ly Project + bang
backend/app/tools/data_analytics/agent/analyst.py	LangChain agent (6 tools)
backend/app/tools/data_analytics/agent/prompts.py	System prompt cho agent
backend/app/tools/data_analytics/analyst_tools/	4 cong cu doc lap
frontend/	Frontend NuxtUI (Vue 3)
frontend/composables/useApi.ts	API client — 9 methods
frontend/pages/tools/data_analytics.vue	Trang Data

Kien truc he thong

Tong quan kien truc

He thong gom 3 dich vu chay trong Docker Compose:



Luong du lieu (Data Flow)

Quy trình xử lý một yêu cầu phân tích dữ liệu:

1. Người dùng truy cập <http://localhost:3000>, Frontend gọi GET `/api/tools` để lấy danh sách tool.
2. Người dùng chọn Data Analytics, Frontend hiển thị ProjectPanel.
3. Người dùng tạo dự án (POST `/project/create`).
4. Người dùng tải file CSV (POST `/project/{id}/upload`). Backend lưu DataFrame vào bộ nhớ (ProjectManager).
5. Người dùng nhập câu hỏi và nhấn "Analyze". Frontend gọi POST `/project/{id}/analyze`.
6. Backend tạo LangChain agent (ChatOllama + 6 tools).
7. Agent thực hiện ReAct loop: gọi Ollama, chọn tool, nhận Observation, lặp lại cho đến khi có kết quả.
8. Agent trả về báo cáo (text) + biểu đồ (Plotly JSON).
9. Frontend hiển thị báo cáo qua LLMReport và biểu đồ qua ChartView (ECharts).

Plugin Registry

Có che `@register_tool()`

Tool Hub cho phép đăng ký công cụ mới bằng một Python decorator. File `backend/app/core/registry.py` định nghĩa có che này:

```

1 import pkgutil
2 import importlib
3 from types import ModuleType
4
5 _tool_registry: dict[str, dict] = {}
6
7
8 def register_tool(
9     name: str,
10    description: str = "",
11    icon: str = "i-heroicons-cube",
12    router: ModuleType | None = None,

```

```

13 ):
14     def decorator(func):
15         _tool_registry[name] = {
16             "name": name,
17             "description": description,
18             "icon": icon,
19             "func": func,
20             "router": router,
21         }
22         return func
23
24     return decorator
25
26
27 def get_all_tools() -> list[dict]:
28     return [
29         {"name": v["name"], "description": v["description"], "icon": v[
30             "icon"]}
31         for v in _tool_registry.values()
32     ]
33
34 def get_tool_router(name: str):
35     entry = _tool_registry.get(name)
36     if entry and entry["router"] is not None:
37         return entry["router"]
38     return None
39
40
41 def discover_tools():
42     package = importlib.import_module("app.tools")
43     for _, modname, ispkg in pkgutil.iter_modules(package.__path__):
44         if not ispkg and modname != "__init__":
45             importlib.import_module(f"app.tools.{modname}")
46         else:
47             importlib.import_module(f"app.tools.{modname}")

```

Trong đó:

- `_tool_registry` — dictionary lưu trữ metadata các tool.
- `@register_tool(name, description, icon, router)` — decorator nhận cho hàm khởi tạo tool.
- `discover_tools()` — tự động quét các module trong `app.tools` bằng `pkgutil.iter_modules`.
- `get_all_tools()` — trả về danh sách tool cho frontend.
- `get_tool_router(name)` — trả về router module của tool.

Khi `app.tools` được import, `discover_tools()` tự động chạy. Mọi tool package có một file `router.py` định nghĩa `APIRouter` riêng.

Cach them tool moi

De them mot tool moi, chi can:

1. Tao thu muc backend/app/tools/<ten-tool>/.
2. Tao file `__init__.py` (rong).
3. Tao file `router.py` voi noi dung:

```

1 from fastapi import APIRouter
2 from app.core.registry import register_tool
3
4 router = APIRouter(prefix="/api/tools/template")
5
6
7 @register_tool(
8     name="template",
9     description="Template tool – copy me to add a new tool",
10    icon="i-heroicons-cube",
11    router=router,
12 )
13 def template_function():
14     pass
15
16
17 @router.get("/hello")
18 async def hello():
19     return {"message": "Hello from template tool!"}

```

Sau do khoi dong lai backend, tool moi xuat hien trong danh sach GET `/api/tools` va tren HubSidebar.

Backend — Data Analytics Tool

API Endpoints

Tool Data Analytics cung cap 9 endpoint REST API trong `router.py`:

Method	Path	Chuc nang
POST	<code>/api/tools/data-analytics/upload</code>	Tai file CSV, tra ve file_id + preview
POST	<code>/api/tools/data-analytics/analyze</code>	Phan tich file CSV (backward compat)
POST	<code>/api/tools/data-analytics/create</code>	Tao slp project (create description)
GET	<code>/api/tools/data-analytics/projects</code>	Danh sach project sa du an
POST	<code>/api/tools/data-analytics/projects/{id}/upload</code>	Tai CSV project (upload)
GET	<code>/api/tools/data-analytics/projects/{id}</code>	Chi tiet project (card)
DELETE	<code>/api/tools/data-analytics/projects/{id}</code>	Xoa slp project
DELETE	<code>/api/tools/data-analytics/projects/{id}/table/{name}</code>	Xoa bang project
POST	<code>/api/tools/data-analytics/projects/{id}/analyze</code>	Phan tich project bang

Gioi han upload: 50 MB (cau hinh trong `config.py`).

Cau hinh (config.py)

```

1 from pydantic_settings import BaseSettings
2 from pathlib import Path
3
4
5 class Settings(BaseSettings):
6     model_name: str = "qwen3:1.7b"
7     ollama_host: str = "http://localhost:11434"
8     cors_origins: str = "http://localhost:3000"
9     upload_dir: str = "backend/app/uploads"
10    max_upload_mb: int = 50
11
12    @property
13    def upload_path(self) -> Path:
14        return Path(self.upload_dir)
15
16    @property
17    def origins_list(self) -> list[str]:
18        return [o.strip() for o in self.cors_origins.split(",")]
19
20    model_config = {"env_file": ".env", "env_file_encoding": "utf-8"}
21
22
23 settings = Settings()

```

Project Manager

project_manager.py dinh nghĩa hai lớp chính:

- **Project:** Lưu trữ project id (UUID), name, description, tables dict (name → DataFrame), table_order list. Các phương thức: add_table (có dedup), remove_table, get_schema_summary, get_context_for_agent.
- **ProjectManager:** Singleton. Các phương thức: create_project, get_project, delete_project, list_projects.

Dữ liệu project được lưu trong bộ nhớ (RAM), sẽ mất khi restart backend. Có thể mở rộng bằng SQLite trong tương lai.

Pydantic Schemas (schemas.py)

Các model Request/Response:

- UploadResponse: file_id, filename, columns, rows, preview
- AnalyzeRequest: question, file_id
- AnalyzeResponse: stats, charts, report
- CreateProjectRequest: name, description
- ProjectResponse: project_id, name, description, tables, table_count
- ProjectAnalyzeRequest: question

- ProjectAnalyzeResponse: project_id, tables_context, report

Agent LangChain

Cach thuc hoat dong

Agent duoc tao trong agent/analyst.py bang ham `create_project_agent(project)`:

1. Tao ChatOllama LLM (model tu config.py, temperature 0.3). **Luu y:** Su dung langchain-ollama.ChatOllama (khong phai langchain_community.llms.Ollama) vi ChatOllama ho tro `.bind_tools()`.
2. Dinh dang SYSTEM_PROMPT voi project context (ten bang, schema, so dong).
3. Dinh nghĩa 6 tool function (xem ben duoi).
4. Tao agent bang `create_agent()` (LangGraph-based API, tra ve CompiledStateGraph).
5. Goi agent voi `agent.invoke({"input": question})`.
6. Ket qua tra ve trong `result["messages"][-1].content` (khong phai `result["output"]`).

Sau cong cu (tools) cua agent

1. **list_tables()**
 - Tra ve danh sach ten bang trong project.
 - Input: Khong co.
 - Output: "Tables: [sales_2024, employees, ...]"
2. **get_table_schema(table_name)**
 - Tra ve ten cot, kieu du lieu, va 3 dong mau.
 - Input: table_name: str
 - Output: "Column 'month' (object), 'revenue' (int64), ..."
3. **get_statistics(table_name)**
 - Thong ke co ban: so dong, so cot, missing values, mo ta so hoc (tu `df.describe()`).
 - Goi `analyst_tools/statistics.py`.
4. **query_table(table_name, query)**
 - Truy van du lieu bang keyword (max, min, average, count, loc, Viet, Anh).
 - Goi `analyst_tools/query_data.py`.
5. **join_tables(table_1, table_2, on_column)**
 - Ghep hai bang theo cot (inner join).

- Bang ket qua duoc luu tam bang ten "joined_<table1>_<table2>".

6. **generate_chart(table_name, chart_type, x_column, y_column)**

- Ve bieu do bang Plotly Express.
- Ho tro: line, bar, scatter, histogram, heatmap (correlation).
- Tu dong phat hien loai bieu do neu chart_type="auto".
- Goi analyst_tools/chart_generator.py.
- Tra ve Plotly JSON de frontend render bang ECharts.

System Prompt

agent/prompts.py chua system prompt dai huong dan agent:

- Su dung ReAct format: Thought / Action / Action Input / Observation / Final Answer.
- Tra loi bang cung ngon ngu voi nguoi dung (tieng Viet hoac tieng Anh).
- Khi can ve bieu do, luon goi generate_chart.
- Khi can so sanh nhieu bang, su dung join_tables truoc.
- Khong duoc tu y tao du lieu, chi phan tich du lieu co san.

Frontend — NuxtUI Components

Cau truc trang

Frontend su dung NuxtUI (Vue 3) voi layout mac dinh gom:

- **layouts/default.vue**: Sidebar ben trai + main content.
- **pages/index.vue**: Trang chu — hien thi danh sach tool (HubToolCard).
- **pages/tools/data-analytics.vue**: Trang Data Analytics — bo cuc 2 panel.

7 Components Data Analytics

Component	Chuc nang
ProjectPanel.vue	Danh sach du an, tao du an moi, chon du an
TableList.vue	Danh sach bang trong du an, xoa bang (nut do khong confirm)
MultiFileUpload.vue	Tai nhieu CSV (drag-drop), kiem tra 50MB, progress bar
FileUpload.vue	Tai file don (backward compat), phan tich nhanh
ChartView.vue	Render bieu do ECharts tu Plotly JSON
StatsTable.vue	Hien thi thong ke (so dong, cot, missing, describe)
LLMReport.vue	Hien thi bao cao AI

API Client (useApi.ts)

```
1 export const useApi = () => {
2   const apiBaseUrl = useRuntimeConfig().public.apiBaseUrl
3
4   const BASE = `${apiBaseUrl}/api/tools/data-analytics`
5
6   async function upload(file: File) {
7     const formData = new FormData()
8     formData.append("file", file)
9     const res = await fetch(`${BASE}/upload`, {
10       method: "POST",
11       body: formData,
12     })
13     if (!res.ok) throw new Error((await res.json()).detail || "Upload
14       failed")
15     return res.json()
16   }
17
18   async function analyze(fileId: string, question: string) {
19     const res = await fetch(`${BASE}/analyze`, {
20       method: "POST",
21       headers: { "Content-Type": "application/json" },
22       body: JSON.stringify({ file_id: fileId, question }),
23     })
24     if (!res.ok) throw new Error((await res.json()).detail || "Analysis
25       failed")
26     return res.json()
27   }
28
29   async function createProject(name: string, description: string) {
30     const res = await fetch(`${BASE}/project/create`, {
31       method: "POST",
32       headers: { "Content-Type": "application/json" },
33       body: JSON.stringify({ name, description }),
34     })
35     if (!res.ok) throw new Error((await res.json()).detail || "Create
36       project failed")
37     return res.json()
38   }
39
40   async function listProjects() {
41     const res = await fetch(`${BASE}/projects`)
42     if (!res.ok) throw new Error("Failed to list projects")
43     return res.json()
44   }
45
46   async function getProject(projectId: string) {
47     const res = await fetch(`${BASE}/project/${projectId}`)
48     if (!res.ok) throw new Error("Project not found")
49     return res.json()
50   }
51 }
```

```
49 async function deleteProject(projectId: string) {
50   const res = await fetch(`${BASE}/project/${projectId}`, { method: "
      DELETE" })
51   if (!res.ok) throw new Error("Delete failed")
52   return res.json()
53 }
54
55 async function uploadToProject(projectId: string, file: File) {
56   const formData = new FormData()
57   formData.append("file", file)
58   const res = await fetch(`${BASE}/project/${projectId}/upload`, {
59     method: "POST",
60     body: formData,
61   })
62   if (!res.ok) throw new Error((await res.json()).detail || "Upload
      failed")
63   return res.json()
64 }
65
66 async function analyzeProject(projectId: string, question: string,
      signal?: AbortSignal) {
67   const res = await fetch(`${BASE}/project/${projectId}/analyze`, {
68     method: "POST",
69     headers: { "Content-Type": "application/json" },
70     body: JSON.stringify({ question }),
71     signal,
72   })
73   if (!res.ok) throw new Error((await res.json()).detail || "Analysis
      failed")
74   return res.json()
75 }
76
77 async function deleteTable(projectId: string, tableName: string) {
78   const res = await fetch(`${BASE}/project/${projectId}/table/${
      encodeURIComponent(tableName)}`, { method: "DELETE" })
79   if (!res.ok) throw new Error((await res.json()).detail || "Delete
      failed")
80   return res.json()
81 }
82
83 return {
84   upload,
85   analyze,
86   createProject,
87   listProjects,
88   getProject,
89   deleteProject,
90   uploadToProject,
91   analyzeProject,
92   deleteTable,
93 }
94 }
```

Data Analytics Page (data-analytics.vue)

Trang chinh bo cuc 2 panel:

- **Panel trai** (288px co dinh): ProjectPanel + TableList.
- **Panel phai**: MultiFileUpload, o nhap cau hoi, nut "Analyze Project", LLMReport.

Tinh nang dac biet:

- Nut "Analyze" bi disable khi cau hoi rong.
- Label nut dong thay doi: "Analyzing..." khi dang xu ly.
- AbortController voi 10-phut timeout tranh loading vo han.

Ollama Integration

Docker Image

ollama/Dockerfile su dung image chinh thuc ollama/ollama va them entrypoint tuy chinh:

```
1 FROM ollama/ollama
2
3 COPY entrypoint.sh /entrypoint.sh
4 RUN chmod +x /entrypoint.sh
5
6 ENTRYPOINT ["/entrypoint.sh"]
```

Entrypoint (auto-pull model)

```
1 #!/bin/sh
2 set -e
3
4 DEFAULT_MODEL=${OLLAMA_DEFAULT_MODEL:-qwen3:4b}
5
6 # Start ollama server in background
7 ollama serve &
8 OLLAMA_PID=$!
9
10 # Wait for server to be ready
11 echo "Waiting for ollama server..."
12 until ollama list 2>/dev/null; do
13     sleep 1
14 done
15 echo "Ollama server ready."
16
17 # Pull default model if not present
18 if [ -n "$DEFAULT_MODEL" ]; then
```

```

19  if ! ollama list 2>/dev/null | grep -q "^${DEFAULT_MODEL%:*}";
    then
20      echo "Model $DEFAULT_MODEL not found. Pulling..."
21      ollama pull "$DEFAULT_MODEL"
22      echo "Model $DEFAULT_MODEL ready."
23  else
24      echo "Model $DEFAULT_MODEL already present."
25  fi
26 fi
27
28 # Bring ollama serve to foreground
29 wait $OLLAMA_PID

```

Entrypoint thực hiện:

1. Chạy ollama serve trong background.
2. Đợi đến khi server sẵn sàng (until ollama list).
3. Kiểm tra model đã được pull chưa; nếu chưa, chạy ollama pull \$OLLAMA_DEFAULT_MODEL.
4. Đưa ollama serve lên foreground bằng wait.

Cấu hình model

Model mặc định: qwen3:1.7b (1.4 GB). Có thể thay bằng qwen3:4b hoặc model khác qua biến môi trường OLLAMA_DEFAULT_MODEL. Điều này được cấu hình trong docker-compose.yml:

```

1 services:
2   ollama:
3     build: ./ollama
4     environment:
5       - OLLAMA_DEFAULT_MODEL=qwen3:1.7b

```

CPU vs GPU

Máy chủ hiện tại sử dụng CPU (AMD Lucienne, không có NVIDIA GPU). Thời gian inference cho qwen3:1.7b khoảng 6.5 phút cho một yêu cầu agent. Khi có GPU, có thể thêm deploy.resources để sử dụng CUDA.

Deployment

Docker Compose (Local)

docker-compose.yml định nghĩa 3 services:

```

1 services:
2   ollama:
3     build: ./ollama
4     ports: ["11434:11434"]

```

```

5   volumes: [ollama_data:/root/.ollama]
6   environment:
7     - OLLAMA_DEFAULT_MODEL=qwen3:1.7b
8
9   backend:
10    build: ./backend
11    ports: ["8000:8000"]
12    depends_on: [ollama]
13    volumes: [./backend:/app]          # hot-reload khi dev
14    environment:
15      - MODEL_NAME=qwen3:1.7b
16      - OLLAMA_HOST=http://ollama:11434
17      - CORS_ORIGINS=http://localhost:3000
18
19  frontend:
20    build: ./frontend
21    ports: ["3000:3000"]
22    depends_on: [backend]
23    environment:
24      - API_BASE_URL=http://localhost:8000
25
26  volumes:
27    ollama_data:

```

Lenh chay:

```

1 docker compose up --build -d
2 docker compose logs -f

```

Docker Compose (Cloud)

Dich vu	Mo ta
Vercel	Frontend (NuxtUI). Build: cd frontend && npm install && npm run build. Output: frontend/.output/public.
Render	Backend (FastAPI). Build: pip install -r backend/requirements.txt. Start: uvicorn app.main:app --host 0.0.0.0 --port 8000.
Ollama	Chay local hoac tren VPS rieng. Khong the deploy len serverless.

Bien moi truong

Bien	Gia tri mac dinh	Dich vu
MODEL_NAME	qwen3:1.7b	Backend
OLLAMA_HOST	http://ollama:11434	Backend
CORS_ORIGINS	http://localhost:3000	Backend
UPLOAD_DIR	backend/app/uploads	Backend
MAX_UPLOAD_MB	50	Backend
API_BASE_URL	http://localhost:8000	Frontend
OLLAMA_DEFAULT_MODEL	qwen3:1.7b	Ollama

Huong dan chay chuong trinh

Cach 1: Docker Compose (khuyen nghi)

Day la cach don gian nhat, khong can cai dat Python hay Node.

1. Cai dat Docker va Docker Compose:

```
1 sudo apt install docker.io docker-compose-v2
2 sudo systemctl start docker
```

2. Chay toan bo he thong:

```
1 cd tool-hub
2 docker compose up --build -d
```

3. Kiem tra trang thai:

```
1 docker compose ps
```

Ca 3 services phai o trang thai Up.

4. Truy cap <http://localhost:3000>.

5. Xem log:

```
1 docker compose logs -f backend
2 docker compose logs -f ollama
```

6. Tat he thong:

```
1 docker compose down
```

Them -v neu muon xoa ca volumes (mat model Ollama).

Cach 2: Thu cong (Local dev)

Buoc 1: Ollama

```
1 # Cai dat Ollama: https://ollama.com
2 ollama serve &
3 ollama pull qwen3:1.7b
```

Buoc 2: Backend

```
1 cd tool-hub/backend
2 python3 -m venv .venv
3 source .venv/bin/activate
4 pip install -r requirements.txt
5 uvicorn app.main:app --host 0.0.0.0 --port 8000 --reload
```

Buoc 3: Frontend

```
1 cd tool-hub/frontend
2 npm install
3 npm run dev
```

Buoc 4: Su dung

1. Mo <http://localhost:3000>.
2. Tao project moi ("Create Project").
3. Tai file CSV (keo tha hoac click).
4. Nhap cau hoi va nhan "Analyze Project".
5. Doi agent xu ly (khoang 6.5 phut tren CPU).

Cach 3: Cloud (Vercel + Render)

1. Push code len GitHub.
2. Tren Render: Tao Web Service, tro den backend/, build command `pip install -r requirements.txt`, start command `uvicorn app.main:app --host 0.0.0.0 --port 8000`.
3. Tren Vercel: Import project, framework la NuxtJS, root directory la frontend/.
4. Cau hinh rewrite trong `vercel.json` de proxy `/api/*` den Render.
5. Ollama can chay rieng (VPS hoac local).

Xu ly loi thuong gap

Loi	Cach xu ly
Backend khong ket noi duoc Ollama	Kiem tra <code>OLLAMA_HOST</code> trong <code>docker-compose.yml</code>
Model chua duoc pull	<code>docker compose exec ollama ollama pull qwen3:1.7b</code>
Frontend loi CORS	Kiem tra <code>CORS_ORIGINS</code> bang voi frontend URL
Upload that bai (file > 50MB)	Tang <code>MAX_UPLOAD_MB</code> trong <code>config.py</code>
Agent tra ve loi	Kiem tra log backend: <code>docker compose logs backend</code>
YAML loi khi docker compose	Kiem tra khoang trang, xoa ky tu dac biet trong <code>.yml</code>

Ket luan

Da hoan thanh

- Tool Hub voi kien truc hybrid (FastAPI + NuxtUI + LangChain + Ollama).
- Co che plugin `@register_tool()` de dang mo rong.
- Data Analytics Tool voi 6 cong tu phan tich du lieu.
- Ho tro da bang (multi-table): agent co the lam viec voi nhieu file CSV trong cung mot du an.
- Docker Compose 3 services chay on dinh.
- Hoi dap song ngu Viet-Anh.
- Bieu do Plotly → ECharts.
- Upload 50 MB.

Huong phat trien

Huong	Mo ta
SQLite persistence	Luu project vao database thay vi in-memory
GPU acceleration	Them CUDA de tang toc inference
Multi-model	Cho phep chon model (Qwen, Llama, Mistral)
Them tool moi	Phan tich PDF, Web scraping, SQL query, ...
Dashboard	Trang tong quan voi nhieu bieu do cung luc
Authentication	Them dang nhap / phan quyen

Thong tin lien quan

Tai lieu	Link
FastAPI	https://fastapi.tiangolo.com
NuxtUI	https://ui.nuxt.com
LangChain	https://python.langchain.com
Ollama	https://ollama.com
Plotly	https://plotly.com/python
ECharts	https://echarts.apache.org
Docker Compose	https://docs.docker.com/compose